S^2MoE: Spiking-based Mixture-of-Experts for Scientific Machine Learning

Shock/Smooth Spatial Specialization for Autoregressive PDE Rollout

A Self-Contained Technical Report

Methods, Background, Equations, Protocol, and Results

Contents

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1 Introduction and problem statement

1.1 What is a PDE surrogate?

Many physical systems (fluids, heat, traffic, phase separation) are described by *partial differential equations* (PDEs): equations relating a field $u(x,t)$ (e.g. velocity, temperature, density) to its

derivatives in space x and time t. Classical numerical solvers integrate a PDE by taking many tiny time steps on a fine grid, which is accurate but expensive. A *neural surrogate* is a neural network trained to imitate the solver: given the field now, predict the field a moment later, and iterate. Because it can take larger steps and run on a GPU, it is potentially much faster.

We consider *autoregressive rollout* surrogates. Let $u^t \in \mathbb{R}^{N_c \times N_x}$ denote the field at discrete time t, sampled on N_x spatial points with N_c physical channels (e.g. N_c=1 for a scalar field, N_c=2 for a two-component system). A rollout surrogate learns a map

$$u^{t+1} = \mathcal{F}_\theta(u^t), \tag{1}$$

and produces a trajectory by feeding its own output back in: $u^t \to u^{t+1} \to u^{t+2} \cdots$. This "closed loop" is powerful but fragile: small per-step errors accumulate and can blow up over a long rollout, so stability is a central concern.

1.2 Why spiking neurons?

Standard neural networks ("artificial neural networks", ANNs) use continuous activations (ReLU, GELU). *Spiking neural networks* (SNNs) instead use neurons that emit discrete 0/1 "spikes" when an internal state crosses a threshold, mimicking biological neurons. Their appeal is *energy*: on neuromorphic hardware, a neuron that does not spike costs (almost) nothing, so sparse spiking can be very cheap. The trade-off is that spikes are discrete and harder to train, and SNNs are usually *less accurate* than ANNs of the same size. Our codebase uses a specific spiking neuron, the *quadratic integrate-and-fire* (QIF) neuron (Section 3.2), for all nonlinearities.

1.3 Why a Mixture-of-Experts?

A *Mixture-of-Experts* (MoE) replaces one network ("dense") by several smaller sub-networks ("experts") plus a small *router* that decides, per input, how much each expert contributes. The intuition is *specialization*: different experts can handle different kinds of input. For a PDE with a moving *shock* (a sharp, nearly discontinuous front) sitting in an otherwise *smooth* field, one might hope that one expert learns to handle the shock and another the smooth background. This report tests exactly that hope.

1.4 Our question and contribution

Does a spiking MoE, by specializing its experts to shock vs. smooth regions, outperform a spiking dense surrogate of comparable size on autoregressive PDE rollout? Our findings:

1. On advective **shock**-forming conservation laws (Burgers, Buckley–Leverett) the MoE beats a parameter-matched spiking dense baseline using *fewer* parameters (∼24%), generalizing to held-out (unseen) PDE parameters.
2. The advantage is caused by *input-dependent routing*, not by merely having more branches: replacing the learned router by uniform weights erases the gain on shock PDEs but not elsewhere.
3. The experts specialize *spatially*: one tracks the moving shock, the other the smooth region.
4. The advantage is *narrow and predictable*: it appears only when there is a moving shock/smooth contrast *and* the dense baseline is capacity-limited. On smooth, elliptic, forced, or "easy" PDEs the MoE ties the dense baseline (it never loses to it in our final configuration).

2 Background concepts (from scratch)

2.1 Spiking neurons: LIF and QIF

A spiking neuron maintains a scalar *membrane potential* v that integrates its input; when v reaches a threshold θ it emits a spike and resets. The simplest model is the *leaky integrate-and-fire* (LIF): $\tau \dot{v} = -v + I$, i.e. v relaxes linearly toward the input I. The *quadratic integrate-and-fire* (QIF) neuron adds a quadratic term:

$$\tau \, \dot{v} = a \, v^2 + \eta + I. \tag{2}$$

This quadratic form is the *normal form of a saddle-node bifurcation* and describes "Type-I" excitability: for $\eta < 0$ there are two fixed points (a stable rest state and an unstable threshold), and as the input I increases they collide and disappear, after which v accelerates to $+\infty$ in finite time — a spike. Intuitively, QIF has a genuine, sharp firing onset (unlike the soft LIF), which is why it is a richer, more nonlinear neuron. We use QIF because it is the neuron in the codebase; Section 3.2 gives our exact discrete-time version.

2.2 Training spikes: surrogate gradients

The spike $s = H(v - \theta)$ (Heaviside step) has zero derivative almost everywhere, so ordinary backpropagation cannot train through it. The standard fix is a *surrogate gradient*: use the true 0/1 spike in the forward pass, but pretend, in the backward pass, that s is a smooth function of v with a bump-shaped derivative peaked at $v = \theta$. This lets gradients flow while keeping spikes discrete at inference.

2.3 Mixture-of-Experts: soft (1991) vs. sparse (2017)

The *original* MoE (Jacobs, Jordan, Nowlan & Hinton, 1991) is *soft*: a gating network outputs a softmax weight for *every* expert, and the output is the weighted sum of *all* experts. Every expert runs on every input. The *modern* MoE (Shazeer et al., 2017; Switch Transformer; Mixtral) is *sparse*: the router picks only the top-k experts ($k \ll E$), so most experts are skipped and compute stays constant as E grows — this is the "efficient MoE" people usually mean. **Our method is the soft, 1991-style MoE** (all experts active, input-dependent weights). We found that sparse top-k routing did *not* help in our setting; only soft gating did. Its efficiency therefore comes from being a *parameter-efficient* multi-expert factorization, not from conditional-compute sparsity — we are explicit about this to avoid over-claiming.

3 Method: the surrogate in full detail

3.1 Rollout backbone

The field enters and leaves through 1D convolutions (kernel size 5, "same" padding), with a *residual* (skip) connection so the network predicts the *change* rather than the full next state (this is standard for rollout and eases stability). With C=48 hidden channels,

$$h = W_1 * u^t \in \mathbb{R}^{C \times N_x}, \qquad u^{t+1} = \mathcal{F}(u^t) = u^t + W_2 * \Phi(h), \tag{3}$$

where $W_1 : N_c \to C$, $W_2 : C \to N_c$ are convolutions and Φ is the block (dense or MoE). The convolution gives each output location a small spatial receptive field; the block Φ is where the spiking nonlinearity and (for the MoE) the routing live.

3.2 QIF spiking neuron (exact discrete form)

We use a one-step Euler discretization of Eq. (2) with a per-timestep reset. For drive x (a channel of h) and membrane state v (carried across rollout steps, initialized to 0):

$$\Delta v = \big(a\, v^2 + \eta + x\big)\, \frac{\Delta t}{\tau}, \tag{4}$$

$$v \leftarrow v + \Delta v, \tag{5}$$

$$s = H(v - \theta) \quad \text{(spike, 0/1)}, \tag{6}$$

$$v \leftarrow v\,(1 - s) + v_{\text{reset}}\, s. \tag{7}$$

Equation (4)–(5) integrates the input with the quadratic self-drive av^2; (6) fires when the threshold is crossed; (7) resets fired units to v_{reset} and leaves the rest unchanged. The learnable parameters (with defaults) are

$$a = 1, \quad \eta = 0, \quad \tau = 2, \quad \Delta t = 1, \quad \theta = 1, \quad v_{\text{reset}} = -1,$$

all trained by gradient descent. The block output is the spike tensor s.

Surrogate gradient. Forward uses the exact step $s = H(v - \theta)$. Backward uses

$$\frac{\partial s}{\partial v} \;\approx\; p\, \exp\big(-w\,|v - \theta|\big), \tag{8}$$

a bump peaked at threshold, with learnable width w and peak p (initialized w=2, p=1).

3.3 Dense block (baseline)

A single QIF feed-forward "expert" with expansion ratio R: project up to width RC, apply the QIF neuron, project back to C:

$$\Phi_{\text{dense}}(h) = W_{\downarrow}\, \text{QIF}\big(W_{\uparrow} h\big), \qquad W_{\uparrow}: C \to RC, \quad W_{\downarrow}: RC \to C. \tag{9}$$

These $W_{\uparrow}, W_{\downarrow}$ are 1×1 convolutions (i.e. per-location linear maps). Our main baseline is **dense-R8** (R=8, hidden width $8C$=384).

3.4 Soft-gated MoE block (our method)

We use E experts, each an up-projection $W_{\uparrow}^{(e)}: C \to RC$, plus a router $W_r: C \to E$, one *shared* QIF neuron, and one down-projection $W_{\downarrow}: RC \to C$. At every spatial location x:

$$g(x) = \text{softmax}\big(W_r\, h(x)\big) \in \mathbb{R}^E \qquad \text{(per-location gate; } \textstyle\sum_e g_e = 1\text{)}, \tag{10}$$

$$m(x) = \sum_{e=1}^{E} g_e(x)\, \big(W_{\uparrow}^{(e)} h(x)\big) \in \mathbb{R}^{RC} \qquad \text{(soft, all-expert mixture)}, \tag{11}$$

$$\Phi_{\text{MoE}}(h) = W_{\downarrow}\, \text{QIF}(m). \tag{12}$$

Key points a first-time reader should note:

- The gate $g(x)$ depends on the *local field content* $h(x)$, so *different spatial locations weight the experts differently* — this is the "routing".

- *All* experts contribute (soft / top-E); we are not skipping any. This is the 1991 MoE.
- The experts differ only in $W_{\uparrow}^{(e)}$; they share the QIF and the down-projection, which keeps the parameter count small.

Our final configuration is **E2-soft-R4**: E=2 experts, ratio R=4.

3.5 Parameter accounting (derivation)

With N_c=1, C=48, kernel 5, count weights+biases:

$$W_1 : 1 \cdot C \cdot 5 + C, \qquad W_2 : C \cdot 1 \cdot 5 + 1,$$
$$W_r : C \cdot E + E, \qquad W_{\downarrow} : RC \cdot C + C, \qquad W_{\uparrow}^{(e)} : C \cdot RC + RC \;\; (\times E).$$

Summing, the MoE and dense totals are

$$P_{\mathrm{MoE}}(E, R) = 2352\, ER + 2304\, R + 49\, E + 583, \qquad P_{\mathrm{dense}}(R) = 4656\, R + 583. \tag{13}$$

Hence **E2-soft-R4** = 28,713 parameters versus **dense-R8** = 37,831 — the MoE is 24% smaller. (For top-k sparse variants one would replace E by k in the *active*, i.e. per-forward, count; we report soft gating where active = total.)

4 Benchmarks: the PDEs and how data is made

All datasets are generated by the lab solver suite, which follows the *CAACT* convention (the same benchmark family used by the spiking-SciML codebase). For each PDE we sweep one physical parameter over 12 trajectories to create *heterogeneity* (a range of regimes), and we evaluate on *held-out* parameter values (interleaved, never seen in training) to test generalization. All fields are resampled to N_x=128 and normalized to zero mean, unit variance using training statistics.

Burgers (advective shock). $\partial_t u + u\,\partial_x u = \nu\,\partial_{xx} u$, viscosity $\nu \in [2, 20] \times 10^{-3}$. The nonlinear advection $u\partial_x u$ steepens the profile into a *moving shock*; the viscosity ν sets its sharpness. Sharp front on a smooth background — the canonical case for our mechanism.

Buckley–Leverett (CAACT form). $\partial_t u = \partial_{xx}\big(2u^2 + \frac{4}{3}u^3\big)$ on $[0, 1]$ with Dirichlet boundaries. A degenerate nonlinear diffusion that develops a sharp front; the heterogeneity parameter is the initial-condition width.

Heat (smooth). $\partial_t u = \alpha\,\partial_{xx} u$, $\alpha \in [5, 50] \times 10^{-3}$, with the analytic $\sin(\pi x)$ Dirichlet mode. Pure diffusion: the field only smooths out; there is *no* sharp feature. This is a negative control: the mechanism should give *no* advantage.

Navier–Stokes 1D (forced Burgers). $\partial_t u + u\,\partial_x u = \nu\,\partial_{xx} u + f(x, t)$ with a random time-varying body force f. The forcing keeps the whole field active, *removing* the quiescent smooth region — a useful contrast to plain Burgers.

LWR traffic (scalar conservation law). $\partial_t \rho + \partial_x\big(\rho(1 - \rho)\big) = 0$. The Lighthill–Whitham–Richards model of traffic; a localized jam steepens into a moving shock. (Implemented here with a Rusanov finite-volume scheme; note this generator is *not* part of the CAACT suite.)

5 Experimental protocol

- **Data.** 12 trajectories per PDE (parameter sweep); train on the trajectories, evaluate free rollout on held-out parameter values. Training horizon T_{tr}=45 steps; evaluation rollout T=55 steps ($> T_{\text{tr}}$, testing extrapolation in time).
- **Loss (teacher forcing).** One-step mean-squared error on the training horizon:

$$\mathcal{L} = \frac{1}{T_{\text{tr}} - 1} \sum_{t=1}^{T_{\text{tr}}-1} \left\| \mathcal{F}(u^t) - u^{t+1} \right\|_2^2. \tag{14}$$

  "Teacher forcing" means each training step is fed the *true* u^t, not the model's own prediction.
- **Optimizer.** AdamW, learning rate 10^{-3} with cosine decay to 10^{-5}, weight decay 10^{-5}, gradient-norm clipping at 1.0, 1500 epochs. (These follow standard neural-PDE practice; a fixed schedule is applied identically to *both* dense and MoE for fairness.)
- **Repeats and significance.** 5 random seeds; the main tables report the mean relative L_2 and a Welch t-statistic for the dense−MoE difference ($|t| > 2 \approx$ significant). **Seed 42 is the canonical reference seed**: released model weights and all figures are produced with seed 42 (its held-out errors, e.g. Burgers 0.097, Buckley 0.064, Heat 0.133, NS 0.186, LWR 0.163, are within seed variation of the 5-seed means reported below).

6 Metrics (defined and motivated)

Relative L_2 error. Accuracy of the free rollout: $\text{relL2} = \|\hat{u} - u\|_2 / \|u\|_2$. A value $\gtrsim 1$ means the prediction is no better than zero (diverged); good surrogates are well below 0.1.

Routing-gain (causal control). To prove that the benefit comes from *input-dependent routing* rather than from simply having several branches, we retrain the MoE with the router *disabled* — fixed uniform gates $g_e \equiv 1/E$ — and measure

$$\text{routing-gain} \;=\; \text{relL2}_{\text{uniform}} - \text{relL2}_{\text{learned}}. \tag{15}$$

A large positive value means the learned router is doing real, causal work.

Specialization correlation. To see *where* experts act, we correlate each expert's time-averaged spatial gate usage $\bar{g}_e(x)$ with the local sharpness $|\partial_x u|(x)$: $\text{corr}\big(\bar{g}_e(x),\, |\partial_x u|(x)\big)$. A strong positive/negative pair means one expert concentrates on the sharp front and the other on the smooth region.

7 Results

Efficiency (Pareto view). On Burgers, spiking dense saturates with width — relative L_2 of 0.175, 0.144, 0.144 at R=2, 4, 8 (params 9.9k, 19.2k, 37.8k) — while the MoE reaches 0.111 at 28.7k, i.e. *below* the dense curve at fewer parameters. Buckley behaves the same. So the MoE is not merely "a bigger network"; it sits above the dense capacity frontier.

Table 1: Main comparison: our E2-soft-R4 MoE (28.7k params) vs. spiking dense-R8 (37.8k params). Held-out evaluation, 5 seeds; lower relative L_2 is better. t is Welch's statistic for (dense − MoE).

Benchmark	character	MoE	dense-R8	t	verdict
Burgers	advective shock	**0.111**	0.144	3.54	**WIN**
Buckley–Leverett	advective shock	**0.075**	0.094	3.33	**WIN**
Heat	smooth (control)	0.133	0.133	−0.34	tie
Navier–Stokes	forced (no quiescence)	0.195	0.210	0.72	tie
LWR	scalar shock (unsat. dense)	0.195	0.197	0.15	tie

Table 2: Causal control and mechanism. Routing-gain isolates the learned router; the specialization correlation shows experts split shock vs. smooth.

Benchmark	routing-gain	t	$\lvert\mathrm{corr}(\bar{g}, \lvert\partial_x u\rvert)\rvert$
Burgers	+0.037	4.48	0.87
Buckley–Leverett	+0.024	4.08	0.42
LWR	+0.041	3.56	0.95
Heat	0.000	0.02	(n/a; smooth)

Interpretation.

1. **The MoE wins only on advective shock PDEs**, and does so with fewer parameters than the dense baseline. On smooth/forced PDEs it ties.
2. **Routing is causal.** Disabling the router (uniform gates) erases the gain on shock PDEs (routing-gain $t > 3.5$) but changes nothing on Heat (routing-gain ≈ 0). So the advantage is the *input-dependent* gating, not the extra branches.
3. **Experts specialize spatially** to shock vs. smooth (strong ± correlations).
4. **Specialization is necessary but not sufficient.** On LWR the router clearly helps the MoE internally (routing-gain +0.041) and specialization is strongest (0.95), yet the MoE only *ties* the dense baseline — because on LWR the dense baseline is not capacity-saturated (adding width keeps helping it). The win over dense therefore needs *both* a routable shock/smooth contrast *and* a capacity-limited dense baseline.

8 Discussion, scope, and honest limitations

- **Where it works.** Moving shock against a quiescent smooth background, with a capacity-limited dense baseline (Burgers, Buckley). This is a *precise, falsifiable* applicability boundary, which we regard as the main contribution rather than a single number.
- **Where it does not.** Smooth diffusion (Heat), elliptic problems, forced flows without a quiescent region (NS), and shock problems that are either easy or where the dense baseline is unsaturated (LWR) — all ties.
- **Soft, not sparse.** Our gains come from the classical soft-gated (1991) MoE; sparse top-k routing did not help. We therefore claim *parameter* efficiency (fewer stored / per-step parameters at equal or better accuracy), *not* the conditional-compute sparsity of modern sparse MoE.

- **Provenance.** Burgers, Buckley, Heat, and NS follow the CAACT lab-solver convention; the LWR generator is our own finite-volume implementation and is flagged as such.